

Solving Oblique Triangles Using the Law of Cosines

Trigonometry Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Apply the Law of Cosines to find missing sides in SAS (Side-Angle-Side) oblique triangles
- Apply the Law of Cosines to find missing angles in SSS (Side-Side-Side) oblique triangles
- Use the alternative form of the Law of Cosines and inverse cosine to determine angle measures including obtuse angles

Problems

1. In triangle ABC, $a = 5$, $b = 7$, and $c = 8$. Use the alternative form of the Law of Cosines to find angle C.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

2. In triangle ABC, $a = 6$, $b = 8$, and $c = 10$. Use the Law of Cosines to determine whether angle C is acute, right, or obtuse.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

3. In triangle ABC, sides $a = 8$, $b = 19$, and $c = 14$. Identify the longest side and find the angle opposite to it.

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

4. In triangle ABC, $a = 10$, $c = 12$, and the included angle $B = 60^\circ$. Use the standard form of the Law of Cosines to find side b.

$$b^2 = a^2 + c^2 - 2accos B$$

5. In triangle ABC, $a = 7$, $b = 9$, and $c = 11$. Find all three angles of the triangle using the Law of Cosines.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$



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6. In triangle ABC, $a = 5$, $b = 6$, and angle $C = 120^\circ$. Find the length of side c .

$$c^2 = a^2 + b^2 - 2ab\cos C$$

7. In triangle ABC, $a = 15$, $b = 20$, and $c = 25$. Find angle A using the alternative form of the Law of Cosines.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

8. Two sides of a triangle measure 13 cm and 17 cm, and the angle between them is 75° . Find the length of the third side and the measure of the largest remaining angle.

$$c^2 = a^2 + b^2 - 2ab\cos C$$

9. A triangular plot of land has sides measuring 120 m, 150 m, and 200 m. Find the largest interior angle of the plot.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

10. In triangle ABC, $a = 9$, $b = 12$, and $c = 16$. First find the obtuse angle using the Law of Cosines, then use the Law of Sines to find a second angle, and finally determine the third angle by subtraction.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} \quad \text{then} \quad \frac{\sin A}{a} = \frac{\sin C}{c}$$



Solving Oblique Triangles Using the Law of Cosines — Answer Key

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Answer Key

1. Answer: $C \approx 75.5^\circ$

- Substitute: $\cos C = (25 + 49 - 64) / (2 \cdot 5 \cdot 7) = 10 / 70 \approx 0.1429$
- Apply inverse cosine: $C = \cos^{-1}(0.1429) \approx 75.5^\circ$

2. Answer: $C = 90^\circ$ (right angle)

- Substitute: $\cos C = (36 + 64 - 100) / (2 \cdot 6 \cdot 8) = 0 / 96 = 0$
- Apply inverse cosine: $C = \cos^{-1}(0) = 90^\circ$, confirming a right angle

3. Answer: $B \approx 116.8^\circ$

- Longest side is $b = 19$; substitute: $\cos B = (64 + 196 - 361) / (2 \cdot 8 \cdot 14) = -101 / 224 \approx -0.4509$
- Apply inverse cosine: $B = \cos^{-1}(-0.4509) \approx 116.8^\circ$

4. Answer: $b \approx 11.1$

- Substitute: $b^2 = 100 + 144 - 2(10)(12)\cos(60^\circ) = 244 - 240(0.5) = 244 - 120 = 124$
- Take square root: $b = \sqrt{124} \approx 11.1$

5. Answer: $A \approx 38.2^\circ$, $B \approx 52.6^\circ$, $C \approx 89.2^\circ$

- Find A: $\cos A = (81 + 121 - 49) / (2 \cdot 9 \cdot 11) = 153 / 198 \approx 0.7727 \rightarrow A \approx 39.5^\circ$; find B similarly using $\cos B = (49 + 121 - 81) / (2 \cdot 7 \cdot 11) \approx 0.6039 \rightarrow B \approx 52.8^\circ$
- Find C: $C = 180^\circ - A - B \approx 180^\circ - 39.5^\circ - 52.8^\circ = 87.7^\circ$; verify with law of cosines directly

6. Answer: $c \approx 9.54$

- Substitute: $c^2 = 25 + 36 - 2(5)(6)\cos(120^\circ) = 61 - 60(-0.5) = 61 + 30 = 91$
- Take square root: $c = \sqrt{91} \approx 9.54$

7. Answer: $A = 36.9^\circ$

- Substitute: $\cos A = (400 + 625 - 225) / (2 \cdot 20 \cdot 25) = 800 / 1000 = 0.8$
- Apply inverse cosine: $A = \cos^{-1}(0.8) \approx 36.9^\circ$

8. Answer: $c \approx 18.0$ cm; largest remaining angle $\approx 70.3^\circ$

- Find c: $c^2 = 169 + 289 - 2(13)(17)\cos(75^\circ) = 458 - 442(0.2588) \approx 458 - 114.4 = 343.6 \rightarrow c \approx 18.0$ cm
- Use Law of Cosines to find the angle opposite $b=17$: $\cos B = (169 + 324 - 289) / (2 \cdot 13 \cdot 18) = 204 / 468 \approx 0.4359 \rightarrow B \approx 64.2^\circ$; angle opposite $a=13$ is $180^\circ - 75^\circ - 64.2^\circ = 40.8^\circ$; largest remaining angle $\approx 64.2^\circ$

9. Answer: $C \approx 93.8^\circ$

- The largest angle is opposite the longest side $c = 200$ m; substitute: $\cos C = (14400 + 22500 - 40000) / (2 \cdot 120 \cdot 150) = -3100 / 36000 \approx -0.0861$



- Apply inverse cosine: $C = \cos^{-1}(-0.0861) \approx 94.9^\circ$

10. Answer: $C \approx 104.5^\circ$, $A \approx 32.1^\circ$, $B \approx 43.4^\circ$

- Find C (obtuse, opposite longest side $c=16$): $\cos C = (81 + 144 - 256)/(2 \cdot 9 \cdot 12) = -31/216 \approx -0.1435 \rightarrow C = \cos^{-1}(-0.1435) \approx 104.5^\circ$
 - Use Law of Sines: $\sin A / 9 = \sin(104.5^\circ) / 16 \rightarrow \sin A = 9 \cdot \sin(104.5^\circ) / 16 \approx 0.5432 \rightarrow A \approx 32.9^\circ$; then $B = 180^\circ - 104.5^\circ - 32.9^\circ \approx 42.6^\circ$
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